Geometry, Triangles & Right-Triangle Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the municipal library, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

Answer: 24. Reading the right triangle diagram, the two perpendicular legs have lengths 6 and 8. The area of the right triangle is $\frac{1}{2} \times 6 \times 8 = 24$.

2 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the coastal aquarium, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

Answer: 15. Reading the right triangle diagram, the two legs have lengths 9 and 12. By the Pythagorean theorem, $AB = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$.

3 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the urban bike-share station, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

Answer: 120. Reading the right triangle diagram, the two perpendicular legs have lengths 10 and 24. The area of the right triangle is $\frac{1}{2} \times 10 \times 24 = 120$.

4 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the agricultural cooperative, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

Answer: 34. Reading the right triangle diagram, the two legs have lengths 16 and 30. By the Pythagorean theorem, $AB = \sqrt{16^2 + 30^2} = \sqrt{1156} = 34$.

5 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the recycling center, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

Answer: 336. Reading the right triangle diagram, the two perpendicular legs have lengths 14 and 48. The area of the right triangle is $\frac{1}{2} \times 14 \times 48 = 336$.

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6 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the figure shown for the roof truss at the neighborhood bookstore, parallel lines ℓ and m are intersected by a transversal line. Based on the angle measure labeled on the diagram, what is the value of x (where x° is the supplementary interior angle)?

Answer: 128. From the transversal diagram, the labeled angle measures 52°. Because consecutive interior angles between parallel lines are supplementary, $x = 180 - 52 = 128$.

7 [Foundational] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the figure shown for the roof truss at the culinary institute, parallel lines ℓ and m are intersected by a transversal line. Based on the angle measure labeled on the diagram, what is the value of x (where x° is the supplementary interior angle)?

Answer: 116. From the transversal diagram, the labeled angle measures 64°. Because consecutive interior angles between parallel lines are supplementary, $x = 180 - 64 = 116$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the orchestra hall, in the circle with center O shown for the courtyard at the campus coffee roastery, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

Answer: 768. Reading the circle diagram, the radius is $r = 48$ and the central angle is 120°. The area of sector AOB is $\frac{120}{360}\pi(48)^2 = 768\pi$, so $k = 768$.

9 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the circle with center O shown for the courtyard at the honeybee apiary, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

Answer: 6. Reading the circle diagram, the radius is $r = 6$ and the central angle is 60°. The area of sector AOB is $\frac{60}{360}\pi(6)^2 = 6\pi$, so $k = 6$.

10 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the local farmers market, in the circle with center O shown for the courtyard at the historic lighthouse, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

Answer: 9. Reading the circle diagram, the radius is $r = 6$ and the central angle is 90°. The area of sector AOB is $\frac{90}{360}\pi(6)^2 = 9\pi$, so $k = 9$.

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11 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the performing arts academy, in the circle with center O shown for the courtyard at the local farmers market, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

Answer: 12. Reading the circle diagram, the radius is $r = 6$ and the central angle is 120°. The area of sector AOB is $\frac{120}{360}\pi(6)^2 = 12\pi$, so $k = 12$.

12 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the coastal aquarium, in the circle with center O shown for the courtyard at the neighborhood hardware store, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

Answer: 6. Reading the circle diagram, the radius is $r = 6$ and the central angle is 60°. The area of sector AOB is $\frac{60}{360}\pi(6)^2 = 6\pi$, so $k = 6$.

13 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the public transit depot, in a calculation for the community theater (case 2), a circle has equation $(x-2)^2 + (y+1)^2 = 16$. What is the radius?

Answer: 4. Applying Area, Volume, Triangles, Circles & Trigonometry principles, $r^2 = 16 \Rightarrow r = 4$ Therefore, the verified correct option is 4.

14 [Intermediate] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the wildlife sanctuary, in the coordinate plane, point A is at $(1, 2)$ and point B is at $(5, 5)$. What is AB?

Answer: 5. $AB = \sqrt{(5-1)^2 + (5-2)^2} = \sqrt{16+9} = 5$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the historic lighthouse, in a calculation for the charter ferry service (case 38), a right triangle has legs of length 7 and 24. What is the hypotenuse?

Answer: 25. $c = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$.

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16 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the botanical conservatory, $\sin(18^\circ) = \cos(x^\circ)$, where $0 < x < 90$. What is the value of x?

Answer: 72. By the complementary angle theorem for right-triangle trigonometry, $\sin(\theta) = \cos(90^\circ - \theta)$. Therefore, $x = 90 - 18 = 72$.

17 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the municipal library layout, the equation $x^2 - 6x + y^2 + 8y = 0$ defines a circle. What is the radius of the circle?

Answer: 5. Complete the square for x and y: $(x - 3)^2 + (y + 4)^2 = 0 + 9 + 16 = 25$. Thus the radius is $r = \sqrt{25} = 5$.

18 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A cylindrical water column at the community theater has a base radius of 5 feet and a height of 7 feet. If the volume of the cylinder is $k\pi$ cubic feet, what is the value of k?

Answer: 175. Using the cylinder volume formula $V = \pi r^2 h$, we have $V = \pi(5)^2(7) = 175\pi$. Therefore, $k = 175$.

19 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the regional bakery, $\sin(21^\circ) = \cos(x^\circ)$, where $0 < x < 90$. What is the value of x?

Answer: 69. By the complementary angle theorem for right-triangle trigonometry, $\sin(\theta) = \cos(90^\circ - \theta)$. Therefore, $x = 90 - 21 = 69$.

20 [Challenging] • [SPR] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the coastal aquarium layout, the equation $x^2 - 12x + y^2 + 14y = -21$ defines a circle. What is the radius of the circle?

Answer: 8. Complete the square for x and y: $(x - 6)^2 + (y + 7)^2 = -21 + 36 + 49 = 64$. Thus the radius is $r = \sqrt{64} = 8$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the botanical conservatory, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

a. 5 ✓

b. 7

c. 4

d. 8

Reading the right triangle diagram, the two legs have lengths 3 and 4. By the Pythagorean theorem, $AB = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

22 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the community theater, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

a. 17

b. 12

c. 13 ✓

d. 15

Reading the right triangle diagram, the two legs have lengths 5 and 12. By the Pythagorean theorem, $AB = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

23 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the regional bakery, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

a. 59

b. 60 ✓

c. 62

d. 23

Reading the right triangle diagram, the two perpendicular legs have lengths 8 and 15. The area of the right triangle is $\frac{1}{2} \times 8 \times 15 = 60$.

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24 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the mountain observatory, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

a. 86

b. 31

c. 83

d. 84 ✓

Reading the right triangle diagram, the two perpendicular legs have lengths 7 and 24. The area of the right triangle is $\frac{1}{2} \times 7 \times 24 = 84$.

25 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the science museum, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

a. 28

b. 19

c. 20 ✓

d. 22

Reading the right triangle diagram, the two legs have lengths 12 and 16. By the Pythagorean theorem, $AB = \sqrt{12^2 + 16^2} = \sqrt{400} = 20$.

26 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the solar microgrid facility, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the length of the hypotenuse AB?

a. 25 ✓

b. 27

c. 35

d. 24

Reading the right triangle diagram, the two legs have lengths 15 and 20. By the Pythagorean theorem, $AB = \sqrt{15^2 + 20^2} = \sqrt{625} = 25$.

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27 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the ceramic studio, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

a. 182

b. 49

c. 179

d. 180 ✓

Reading the right triangle diagram, the two perpendicular legs have lengths 9 and 40. The area of the right triangle is $\frac{1}{2} \times 9 \times 40 = 180$.

28 [Foundational] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In right triangle ABC shown for a support frame at the publishing house, $\angle C = 90^\circ$. Based on the leg lengths labeled in the figure, what is the area of ΔABC in square units?

a. 209

b. 210 ✓

c. 212

d. 47

Reading the right triangle diagram, the two perpendicular legs have lengths 12 and 35. The area of the right triangle is $\frac{1}{2} \times 12 \times 35 = 210$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the wildlife sanctuary, in the circle with center O shown for the courtyard at the environmental club, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 324 ✓

b. 162

c. 72

d. 360

Reading the circle diagram, the radius is $r = 36$ and the central angle is 90°. The area of sector AOB is $\frac{90}{360}\pi(36)^2 = 324\pi$, so $k = 324$.

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30 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the charter ferry service, in the circle with center O shown for the courtyard at the neighborhood hardware store, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 36

b. 27

c. 24 ✓

d. 12

Reading the circle diagram, the radius is $r = 12$ and the central angle is 60°. The area of sector AOB is $\frac{60}{360}\pi(12)^2 = 24\pi$, so $k = 24$.

31 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the textile workshop, in the circle with center O shown for the courtyard at the aquatic center, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 168

b. 144 ✓

c. 72

d. 48

Reading the circle diagram, the radius is $r = 24$ and the central angle is 90°. The area of sector AOB is $\frac{90}{360}\pi(24)^2 = 144\pi$, so $k = 144$.

32 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the geology lab, in the circle with center O shown for the courtyard at the archery range, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 432 ✓

b. 216

c. 72

d. 468

Reading the circle diagram, the radius is $r = 36$ and the central angle is 120°. The area of sector AOB is $\frac{120}{360}\pi(36)^2 = 432\pi$, so $k = 432$.

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33 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the carpentry workshop, in the circle with center O shown for the courtyard at the wind energy cooperative, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 48

b. 36 ✓

c. 18

d. 24

Reading the circle diagram, the radius is $r = 12$ and the central angle is 90°. The area of sector AOB is $\frac{90}{360}\pi(12)^2 = 36\pi$, so $k = 36$.

34 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the historic lighthouse, in the circle with center O shown for the courtyard at the carpentry workshop, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 108 ✓

b. 54

c. 36

d. 126

Reading the circle diagram, the radius is $r = 18$ and the central angle is 120°. The area of sector AOB is $\frac{120}{360}\pi(18)^2 = 108\pi$, so $k = 108$.

35 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the community orchard, in the circle with center O shown for the courtyard at the marine research vessel, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 48

b. 120

c. 99

d. 96 ✓

Reading the circle diagram, the radius is $r = 24$ and the central angle is 60°. The area of sector AOB is $\frac{60}{360}\pi(24)^2 = 96\pi$, so $k = 96$.

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36 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the environmental club, in the circle with center O shown for the courtyard at the national park visitor center, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 51

b. 48 ✓

c. 24

d. 60

Reading the circle diagram, the radius is $r = 12$ and the central angle is 120°. The area of sector AOB is $\frac{120}{360}\pi(12)^2 = 48\pi$, so $k = 48$.

37 [Intermediate] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the neighborhood hardware store, in the circle with center O shown for the courtyard at the community orchard, the radius and central angle of shaded sector AOB are labeled. If the area of shaded sector AOB is $k\pi$, what is the value of k?

a. 54 ✓

b. 27

c. 36

d. 72

Reading the circle diagram, the radius is $r = 18$ and the central angle is 60°. The area of sector AOB is $\frac{60}{360}\pi(18)^2 = 54\pi$, so $k = 54$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the honeybee apiary, in a calculation for the wildlife sanctuary (case 36), what is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. **96 cm²** ✓

c. 128 cm²

d. 144 cm²

A cube has 6 congruent square faces: $SA = 6s^2 = 6(4^2) = 6(16) = 96$ cm^2.

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39 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the carpentry workshop, in a circle, a central angle of 60° intercepts what fraction of the entire circumference?

a. $\frac{1}{4}$

b. $\frac{1}{5}$

c. $\frac{1}{6}$ ✓

d. $\frac{1}{8}$

$\frac{60^\circ}{360^\circ} = \frac{1}{6}$.

40 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the community orchard, in a calculation for the textile workshop (case 39), what is the area of a parallelogram with base 12 cm and height 5 cm?

a. 30 cm²

b. **60 cm²** ✓

c. 90 cm²

d. 120 cm²

Area = base × height = $12 \times 5 = 60$ cm^2.

41 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a model for the local farmers market, a rectangle has perimeter 36 cm and length = twice width. What is the area in cm²?

a. 36

b. 48

c. 64

d. 72 ✓

$2(2w + w) = 36 \Rightarrow 6w = 36 \Rightarrow w = 6, l = 12$. Area $= 6 \times 12 = 72$ cm^2

42 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the orchestra hall layout, the equation $x^2 - 12x + y^2 + 10y = 39$ defines a circle. What is the radius of the circle?

a. 11

b. 12

c. 10 ✓

d. 100

Complete the square for x and y: $(x - 6)^2 + (y + 5)^2 = 39 + 36 + 25 = 100$. Thus the radius is $r = \sqrt{100} = 10$.

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43 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

A cylindrical water column at the sports complex has a base radius of 6 feet and a height of 12 feet. If the volume of the cylinder is $k\pi$ cubic feet, what is the value of k?

a. 438

b. 432 ✓

c. 144

d. 72

Using the cylinder volume formula $V = \pi r^2 h$, we have $V = \pi(6)^2(12) = 432\pi$. Therefore, $k = 432$.

44 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In a right triangle survey at the charter ferry service, $\sin(42^\circ) = \cos(x^\circ)$, where $0 < x < 90$. What is the value of x?

a. 48 ✓

b. 42

c. 138

d. 58

By the complementary angle theorem for right-triangle trigonometry, $\sin(\theta) = \cos(90^\circ - \theta)$. Therefore, $x = 90 - 42 = 48$.

45 [Challenging] • [MCQ] • Target: Area, Volume, Similar Triangles, Circles & Right-Triangle Trig

In the xy-plane for the maple syrup farm layout, the equation $x^2 - 6x + y^2 + 6y = 7$ defines a circle. What is the radius of the circle?

a. 25

b. 6

c. 7

d. 5 ✓

Complete the square for x and y: $(x - 3)^2 + (y + 3)^2 = 7 + 9 + 9 = 25$. Thus the radius is $r = \sqrt{25} = 5$.

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